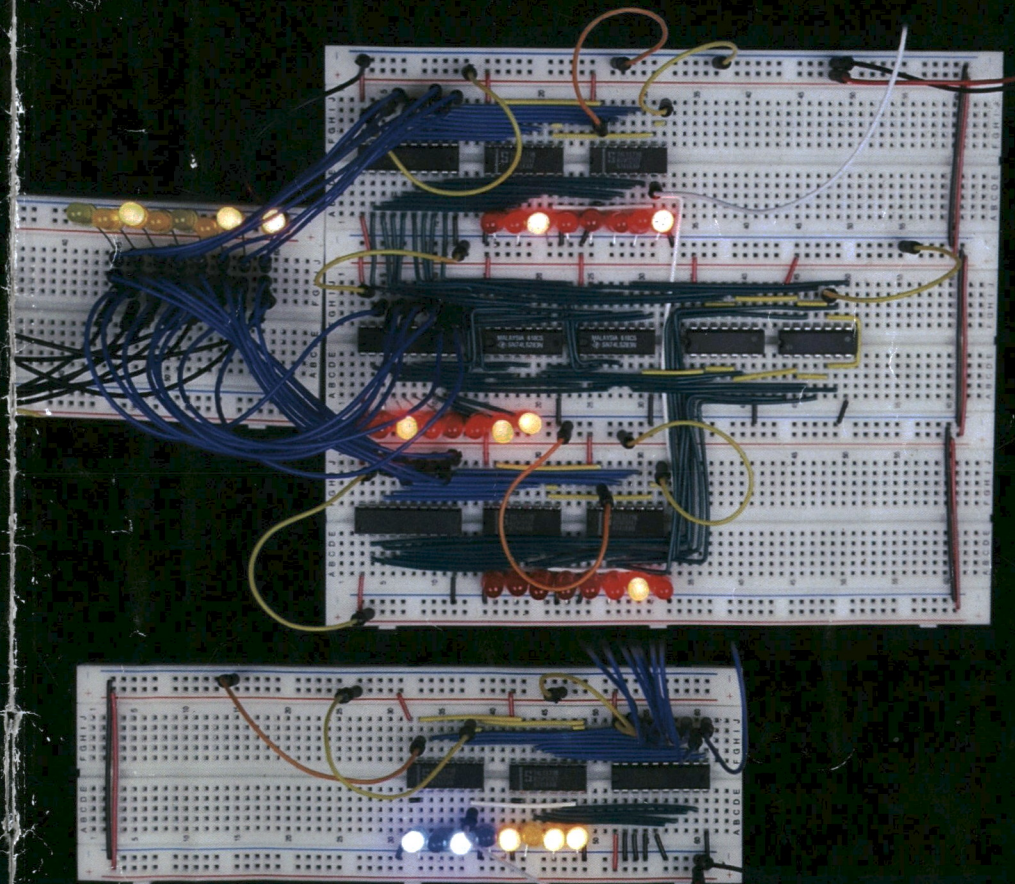


8-bit breadboard computer

Kit 2 of 4

Registers and ALU

www.eater.net
youtube.com/beneater



Ben Eater

Welcome!

This is the second kit in a set of four that, along with my series of videos on YouTube, will help you develop a deep understanding of how programmable CPUs really work. I hope when you're done, you'll know enough to extend the computer design and make it your own.

With this second kit, you'll build the computer's registers and arithmetic and logic unit (ALU).

Most CPUs have a number of registers which store small amounts of data that the CPU is processing. In our breadboard CPU, we'll build three 8-bit registers: A, B, and IR. The A and B registers are general-purpose registers. IR (the instruction register) works similarly, but we'll only use it for storing the current instruction that's being executed.

The arithmetic logic unit (ALU) part of a CPU is usually capable of performing various arithmetic, bitwise, and comparison operations on binary numbers. In our simple breadboard CPU, the ALU is just able to add and subtract. It's connected to the A and B registers and outputs either the sum of $A+B$ or the difference of $A-B$.

Where are the instructions to build this kit?

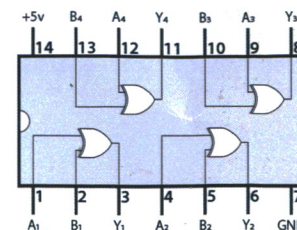
Surprise! This booklet doesn't actually contain the full instructions for building the kit. This kit is a companion to my series of YouTube videos. You should watch the videos and follow along yourself. I hope that by *not* having a detailed "paint-by-numbers" instruction book, you'll learn more about how the circuits really work:

The video instructions for this kit are at
www.eater.net/8bit/registers
www.eater.net/8bit/alu

Integrated circuit quick reference

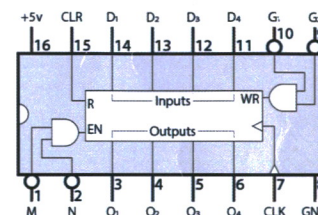
74LS86: XOR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0



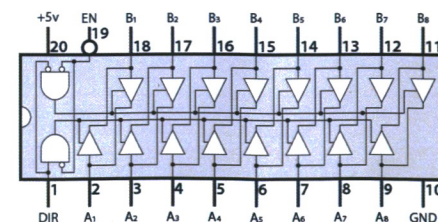
74LS173: 4-bit D register

This register stores 4 bits. You write data to it by holding G1 and G2 low and pulsing the clock.



74LS245: 8-bit bus transceiver

This allows 8-bit data from A to B or B to A depending on the DIR input. The EN input disables the device so that A and B are isolated.



74LS283: 4-bit binary adder

This chip performs addition of two 4-bit binary numbers. $A+B=\Sigma$. Co and C4 are carry in and carry out for chaining multiple adders together

